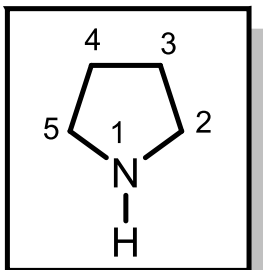
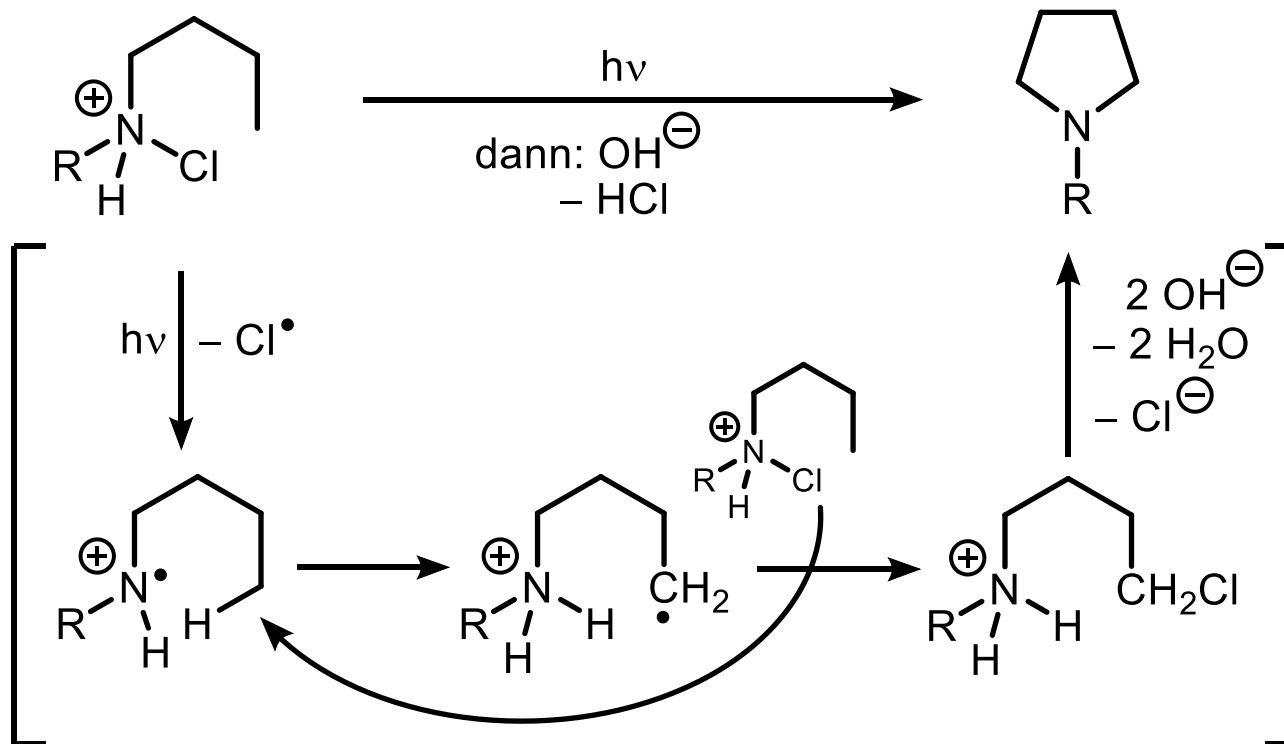


4.4.4. Pyrrolidin



Hofmann-Löffler-Freytag-Umlagerung

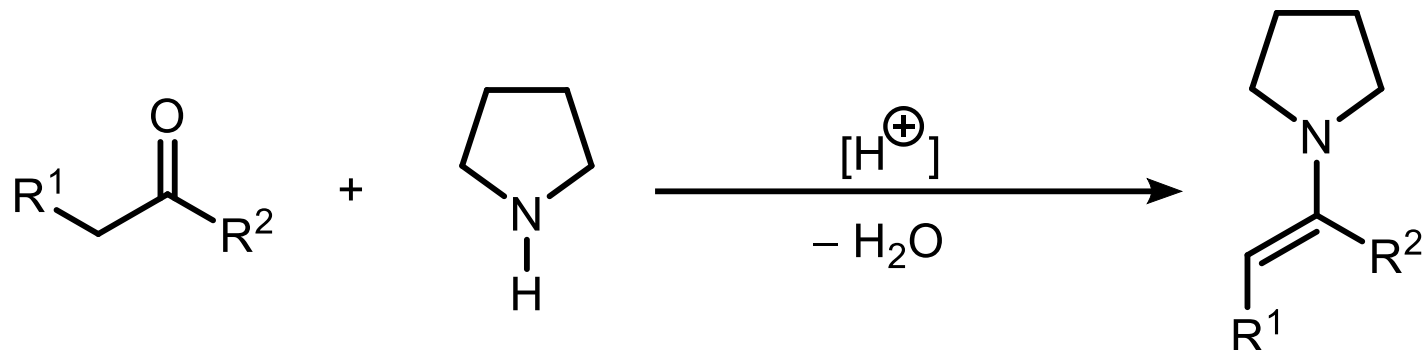


4.4.4. Pyrrolidin

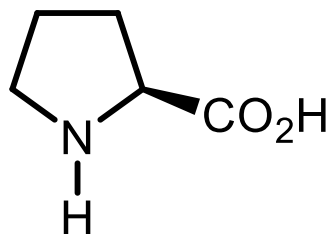
Reaktionen, Vorkommen

Pyrrolidino-Enamine

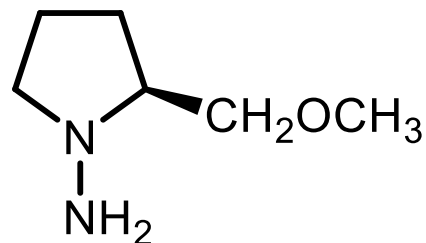
55



Vorkommen der Pyrrolidine und Verwendung



(S)-(-)-Prolin



(S)-1-Amino-2-(methoxymethyl)pyrrolidin
(SAMP)

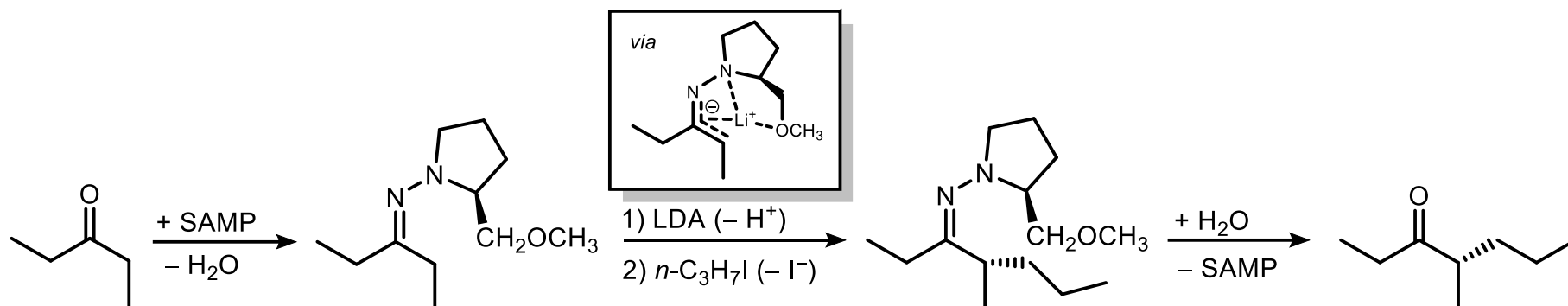
aus enantiomerenreinem Prolin
hergestellt ("ex chiral pool" ECP)

4.4.4. Pyrrolidin

Enders' Methode zur Erzeugung von chiralen Azaenolaten

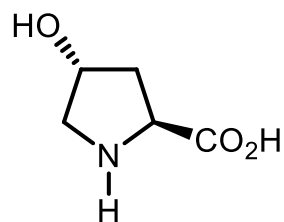
SAMP und RAMP aus (S)- bzw. (R)-Prolin

56

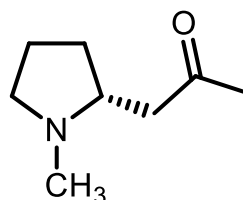


Alkaloide

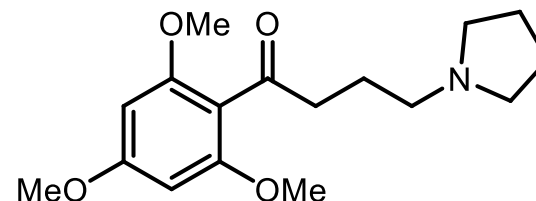
Pharmaka



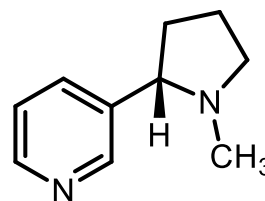
4-Hydroxyprolin
(im Collagen)



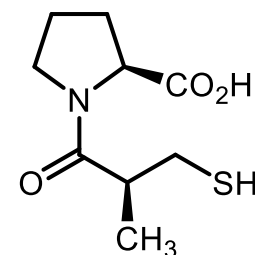
Hygrin
(aus der Coca-Pflanze)



Buflomedil
(Vasodilator)



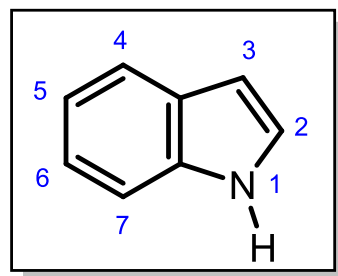
Nicotin



Captopril
(Antihypertensivum)

4.5. Indol

4.5.1. Struktur, Eigenschaften



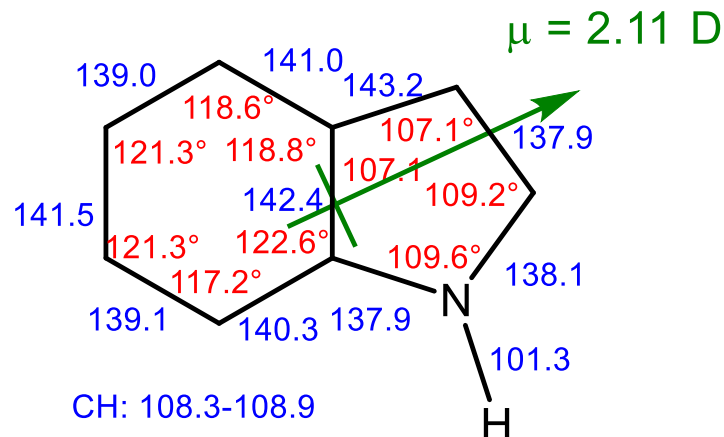
$pK_S = 16.2$ (in Wasser)

$pK_S = 21.0$ (in DMSO)

$pK_S (\text{Indol-H}^+) = -3.5$

$pK_B (\text{Indol}) = 17.5$

Smp. 232 °C



$^1\text{H-NMR}$ (Aceton- d_6)

δ (ppm)

H-1 10.12

H-2 7.27

H-3 6.45

H-4 7.55

$^{13}\text{C-NMR}$ (CDCl_3)

δ (ppm)

C-2 123.7 C-6 119.0

C-3 101.8 C-7 110.4

C-4 119.9 C-3a 127.0

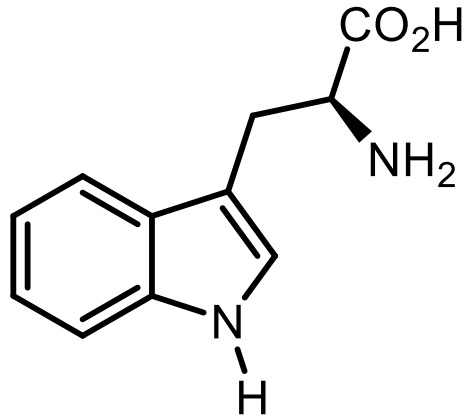
C-5 121.1 C-7a 134.8

Vorkommen: Steinkohleteer, viele Naturstoffe (Alkaloide), Aminosäure Tryptophan

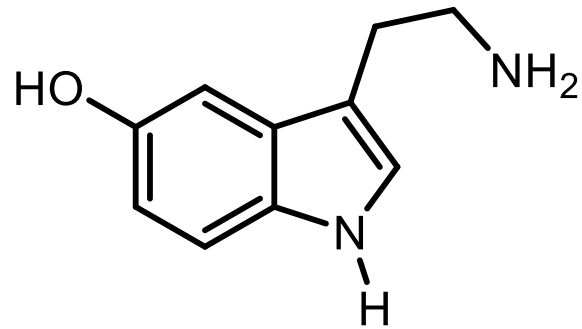
Geruch: blumig in verdünnten Lösungen, fäkalienartig in konzentrierten Lösungen

Skatol (3-Methylindol): Abbauprodukt von Tryptophan, typischer Fäkaliengeruch

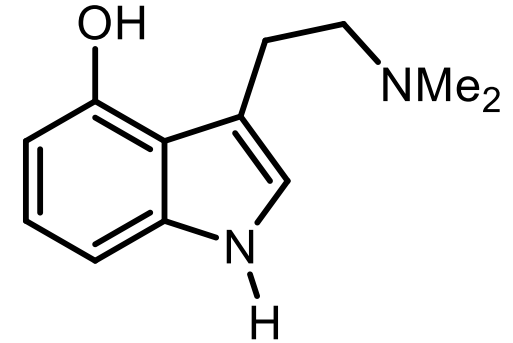
4.5.2. Vorkommen



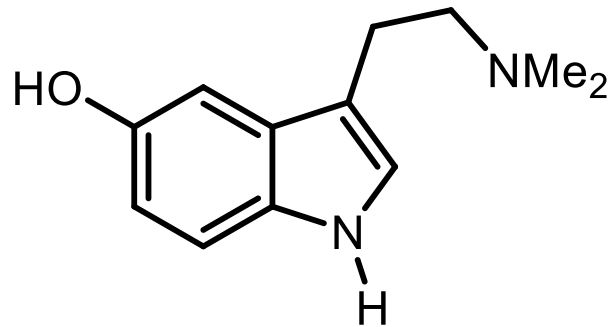
L-Tryptophan (Trp, W)
proteinogene Aminosäure



Serotonin
(Neurotransmitter)



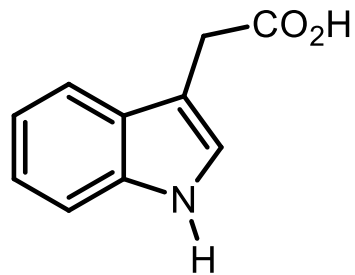
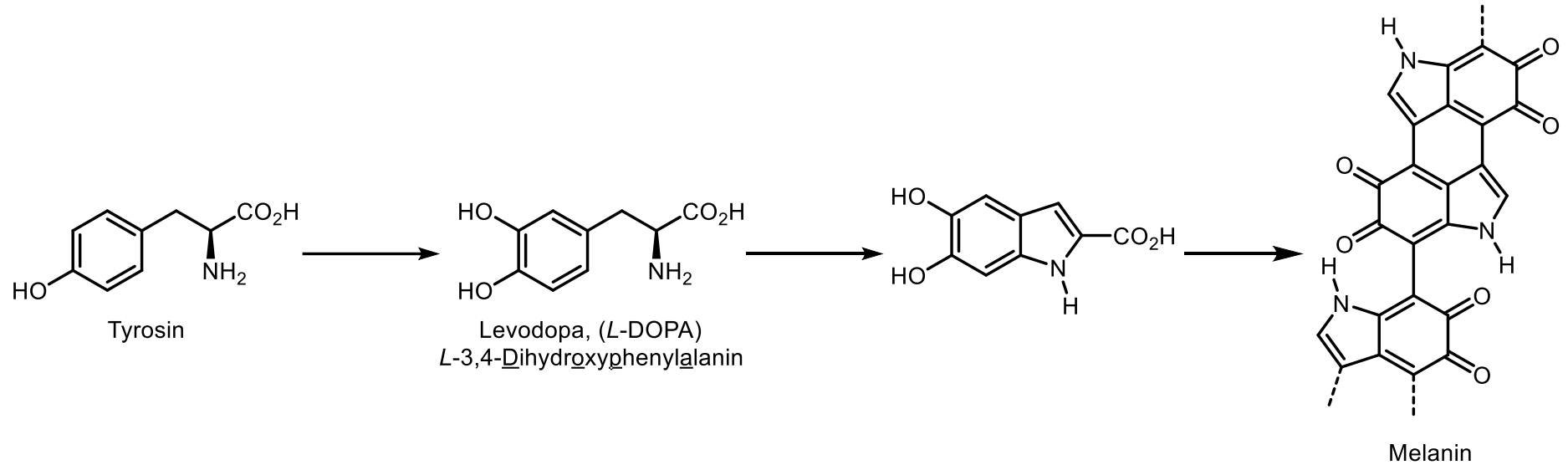
Psilocin
Psilocybin (Phosphorsäureester)



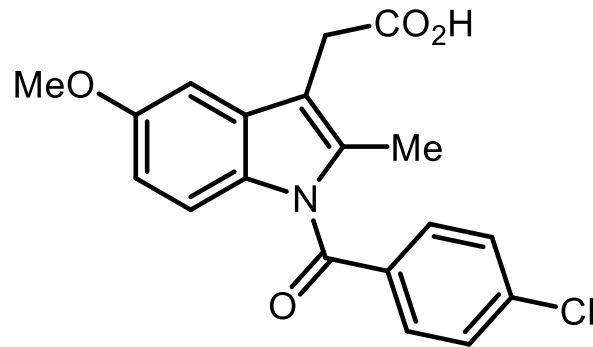
Bufotenin
(Krötengift)

Psilocybin, Psilocin: Indol-Alkaloide aus dem mexikanischen Rauschpilz "Teonanácatl" ("Gottesfleisch"), halluzinogen; die orale Aufnahme ruft Farbvisionen, ein Gefühl der Bewusstseinerweiterung, auch der Persönlichkeitsspaltung und eine stark erhöhte Lichtempfindlichkeit hervor. Ca. 1 % der Wirkung von LSD. Mechanismus: Serotonin-Rezeptor-Agonisten (Drug and Alcohol Dependence **1998**, 189).

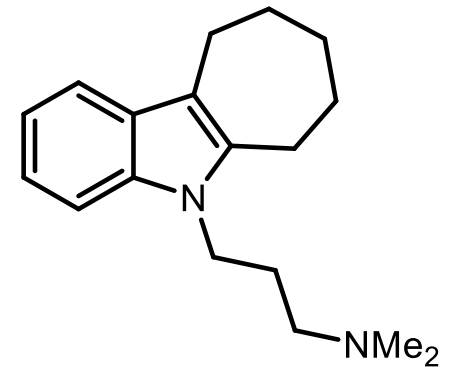
4.5.2. Vorkommen Indolderivate



3-Indolyllessigsäure
(Phytohormon, stimuliert
Pflanzenwachstum)



Indometacin
(Antiphlogisticum)

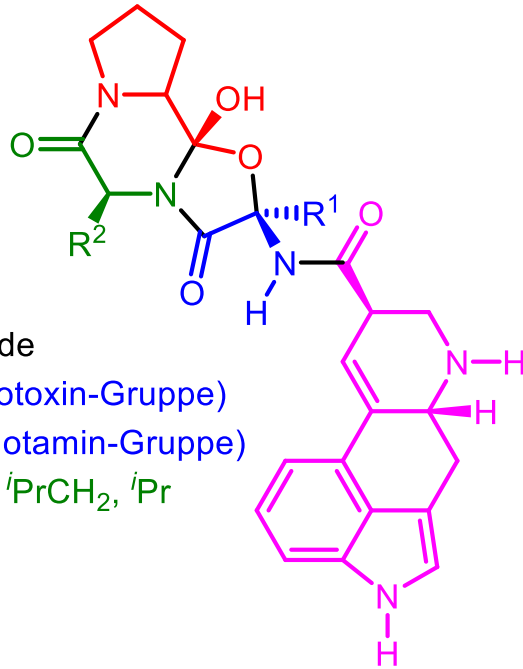


Iprindol
(Antidepressivum)

4.5.2. Vorkommen

Vorkommen (Indolderivate)

Mutterkorn-Alkaloide aus
Claviceps purpurea



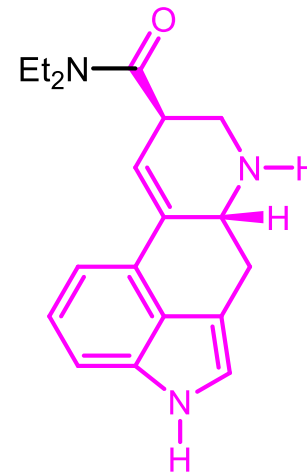
Ergot-Alkaloide

$R^1 = i\text{Pr}$ (Ergotoxin-Gruppe)

$R^1 = \text{Me}$ (Ergotamin-Gruppe)

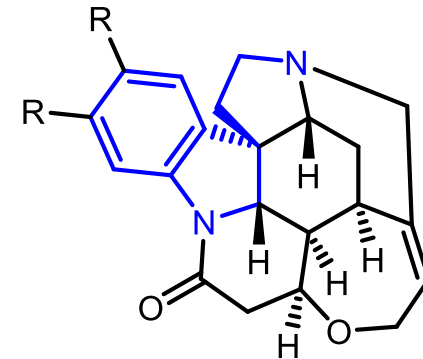
$R^2 = \text{CH}_2\text{Ph}, i\text{PrCH}_2, i\text{Pr}$

Synthetische Derivate
(Halluzinogene)



Lysergsäurediethylamid
(LSD)

Strychnos-Alkaloide

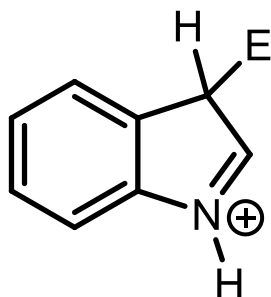


$R = \text{H}$ (Strychnin); $R = \text{OMe}$ (Brucin)
aus Krähenaugenbaum
Strychnos nux vomica
(Brechnuß)

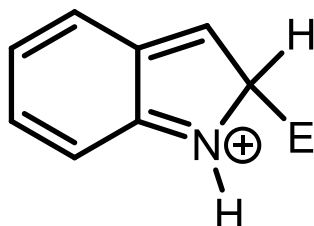
4.5.3. Reaktivität

S_EAr : σ -Komplexe

57



benzoider Aromat
intakt $\Rightarrow$ günstig



ortho-chinoid
 $\Rightarrow$ höhere Energie

$\Rightarrow S_EAr$ an 3-Position bevorzugt

wenn 3-Position blockiert: 2-Position > 6-Position

58

3-Chlorierung mit SOCl_2 od. aq. NaOCl

3-Bromierung mit NBS

3-Sulfonierung mit Pyridin- SO_3

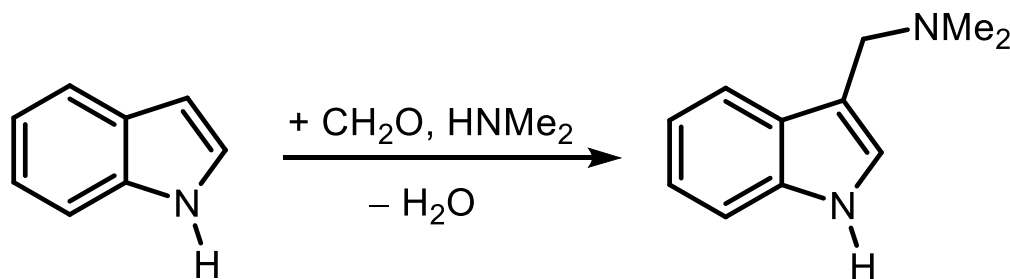
3-Formylierung nach Vilsmeier-Haack

3-Acetylierung mit Ac_2O

Gramin-Synthese mit $\text{H}_2\text{CO}/\text{HNMe}_2/\text{H}^+$

N-Alkylierung nach Deprotonierung mit NaH

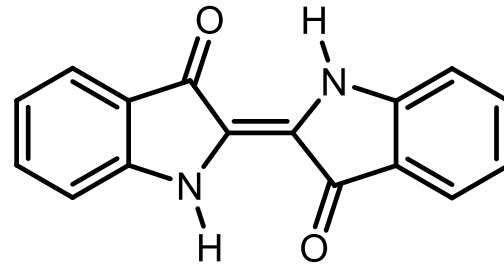
Gramin-Synthese



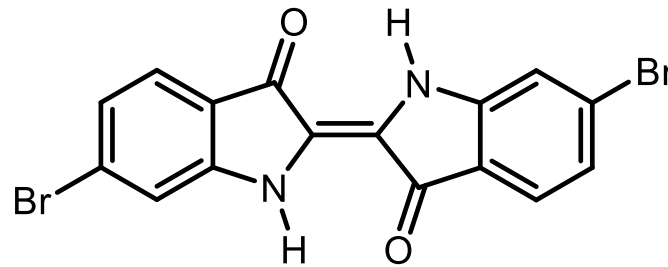
4.5.3. Reaktivität Indigo



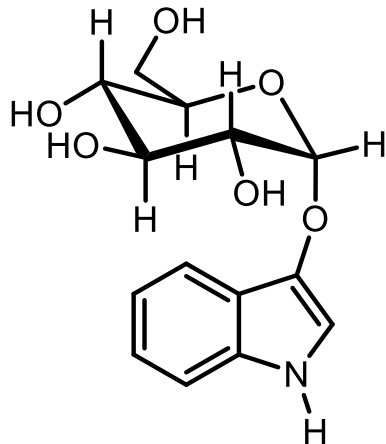
Adolf von-Baeyer
(1835-1917)



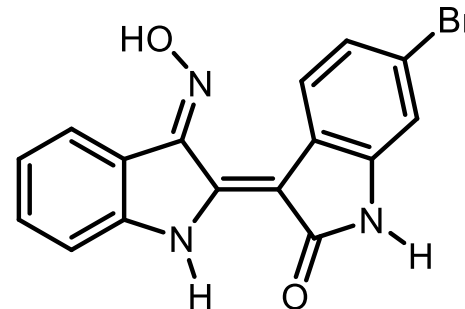
Indigo (blau)



6,6'-Dibromindigo (purpur; aus
Purpurschnecke *hexaplex trunculus*)



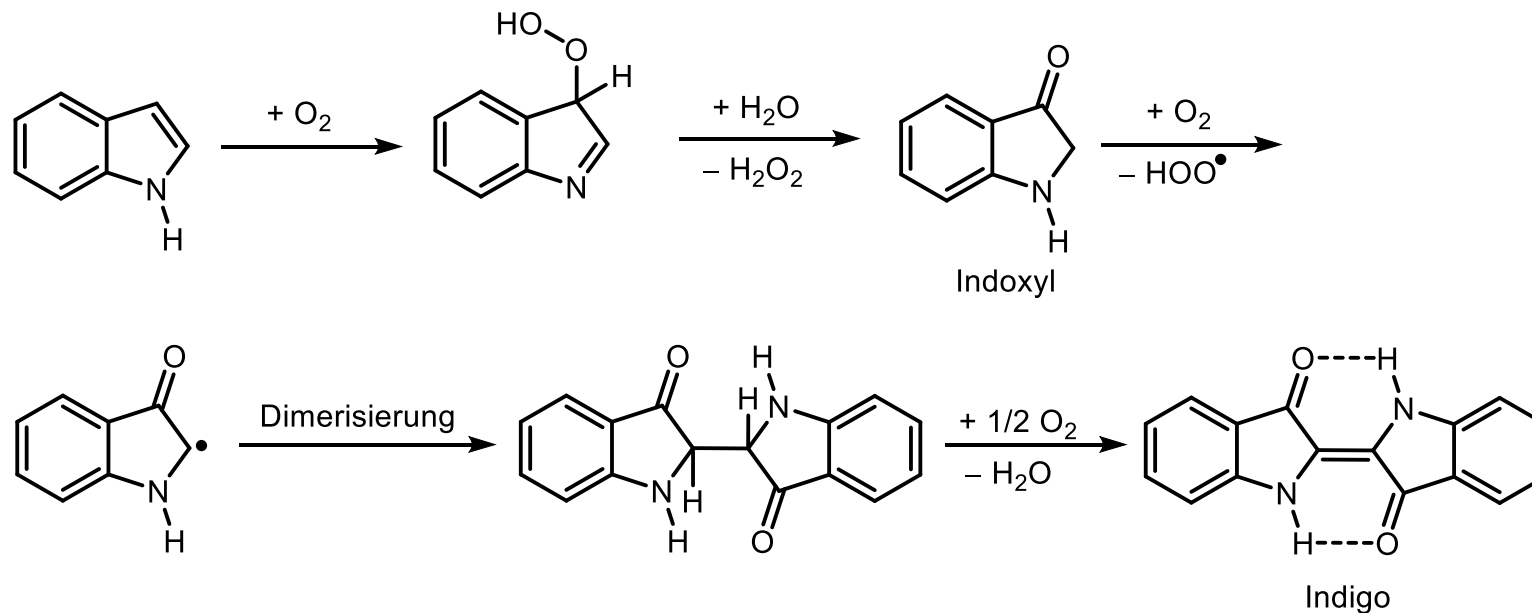
Indican
(aus Indigopflanze
indigofera tinctoria)



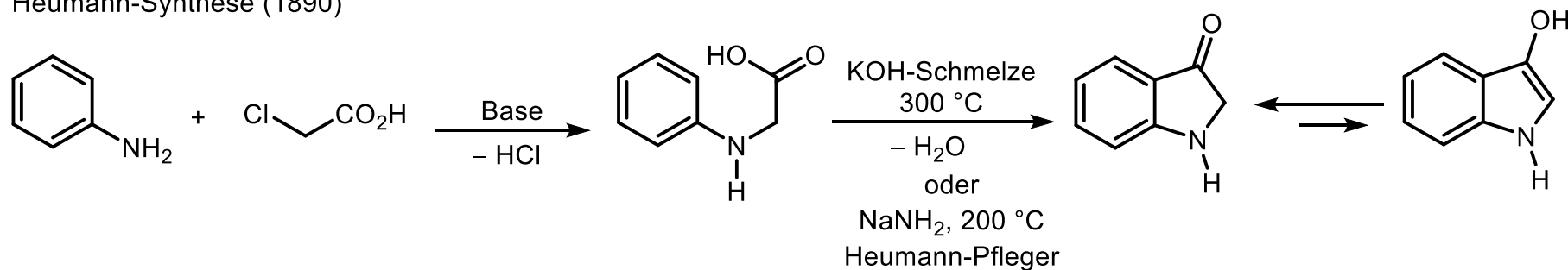
6-Bromindirubin-3'-oxim
(zur Konservierung von
Stammzellen)

4.5.3. Reaktivität Indigobildung

59

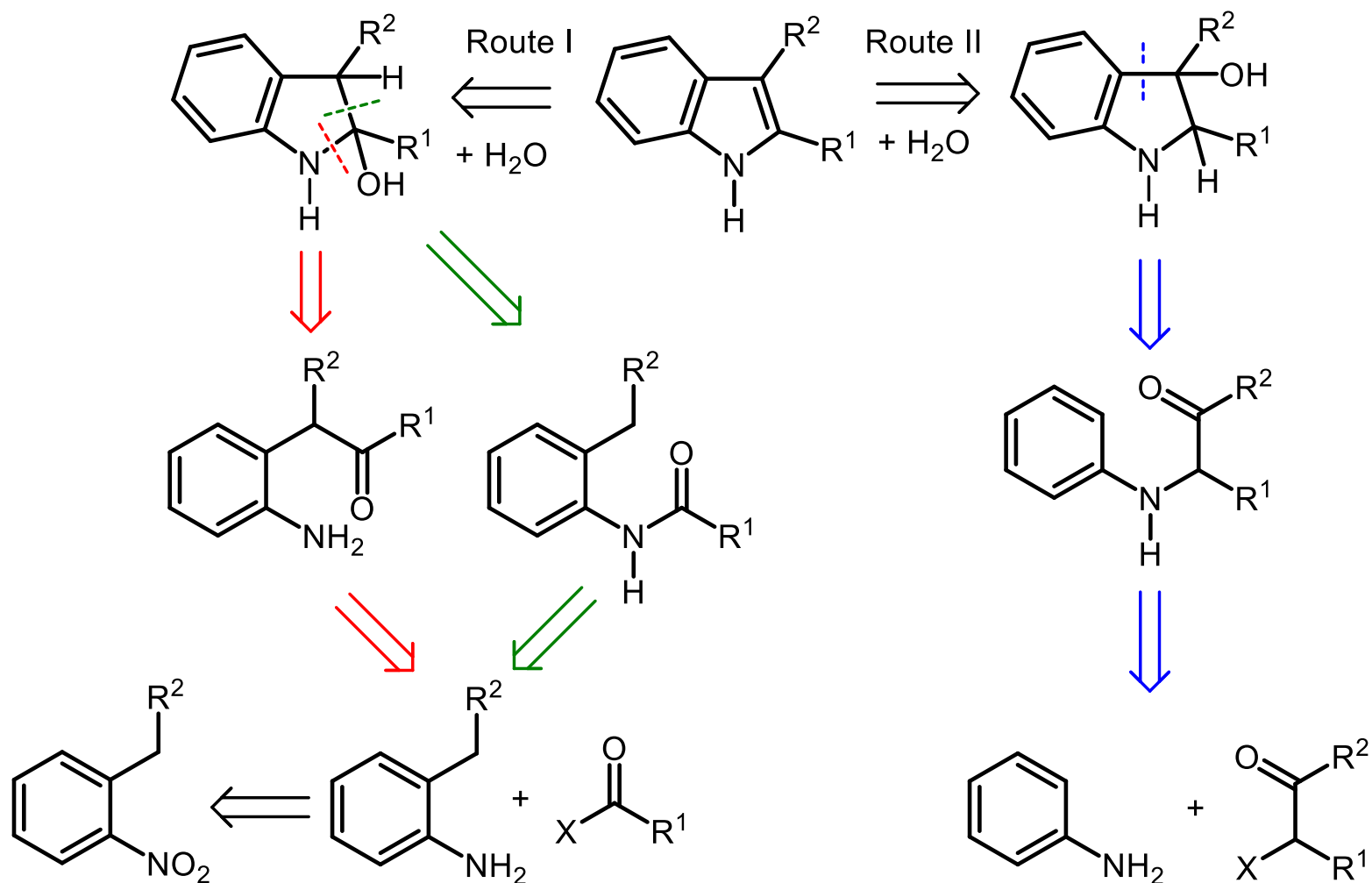


Heumann-Synthese (1890)



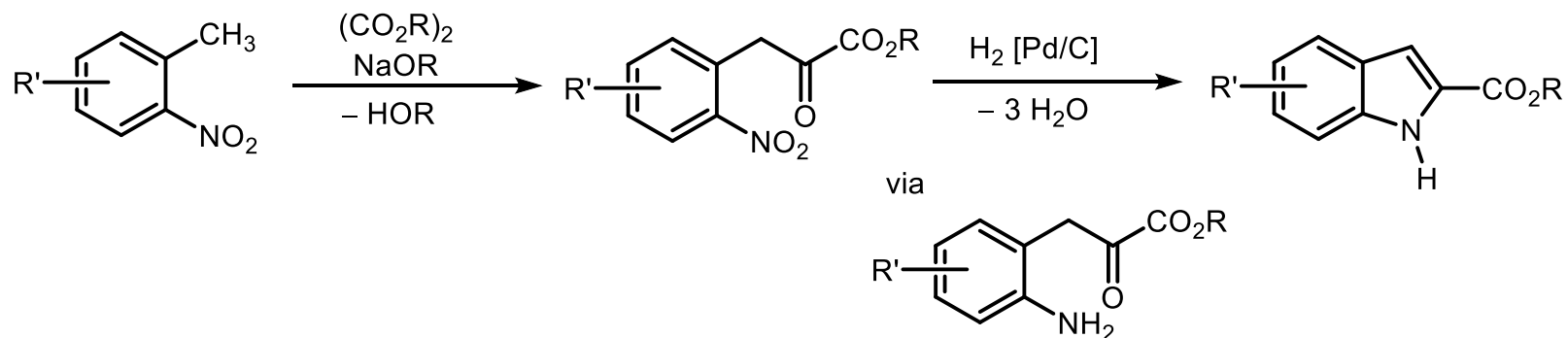
4.5.4. Ringaufbauende Synthesen

Retrosynthetische Analyse

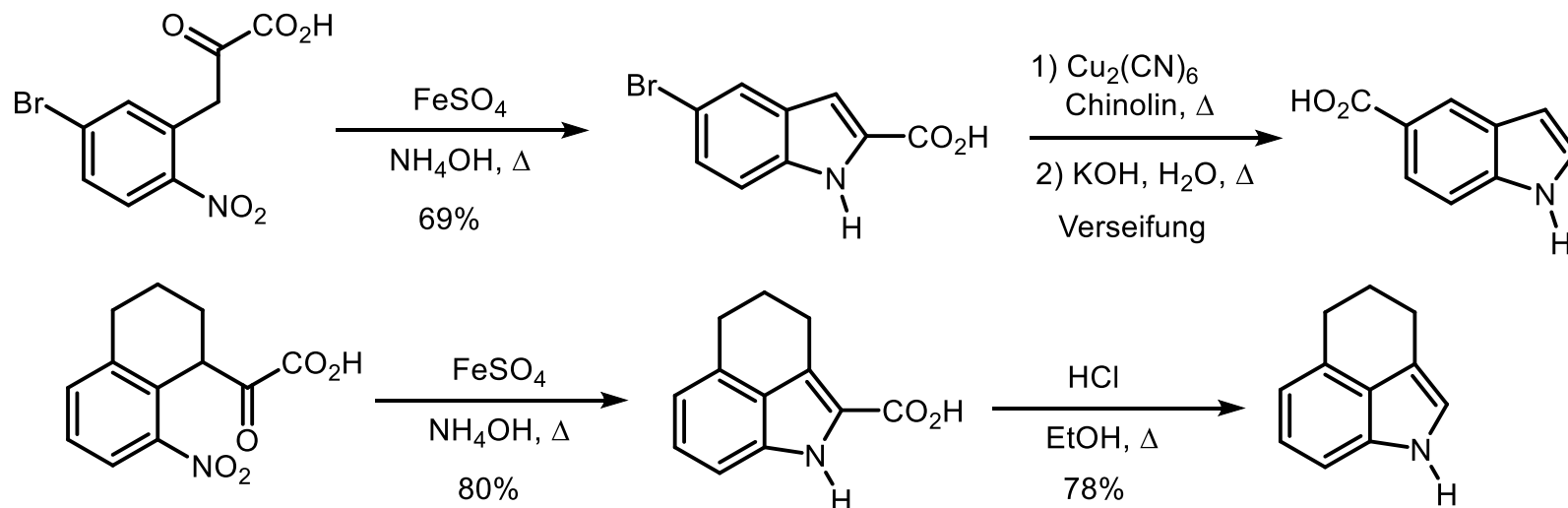


4.5.4. Ringaufbauende Synthesen

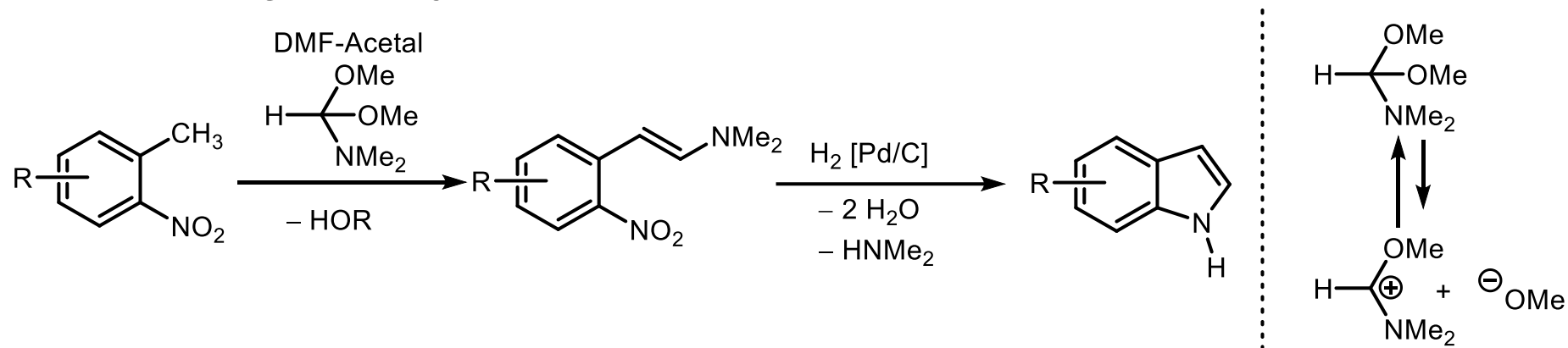
Reissert-Synthese



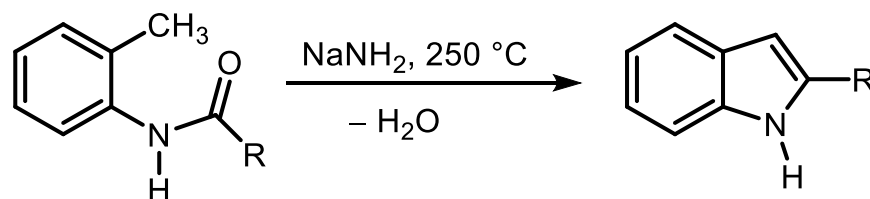
Rosenmund-von Braun-Cyanierung/Decarboxylierung



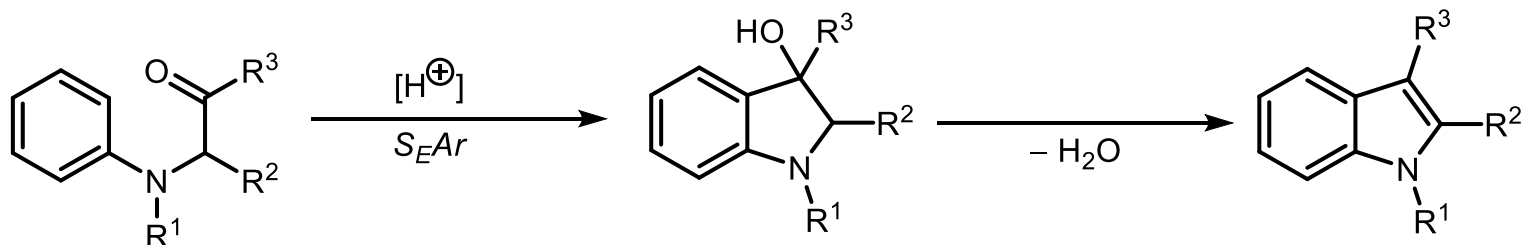
4.5.4. Ringaufbauende Synthesen Batcho-Leimgruber-Synthese



Madelung-Synthese



Bischler-Synthese

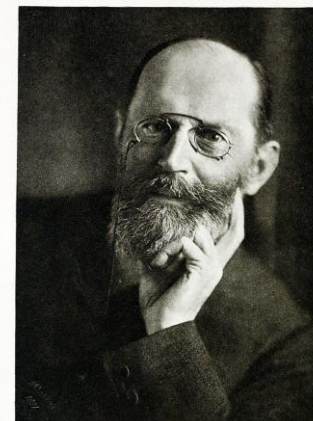
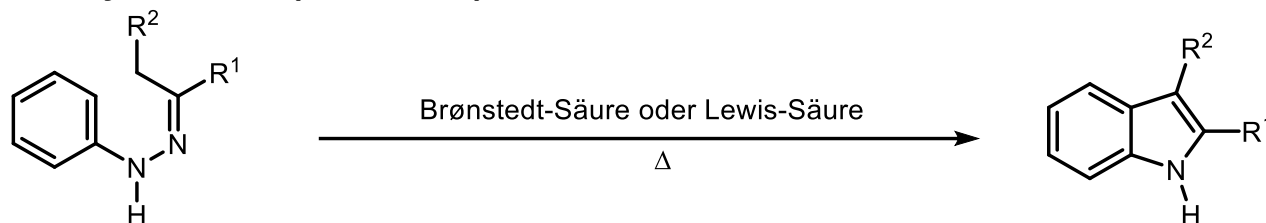


4.5.4. Ringaufbauende Synthesen

Fischer-Synthese

60

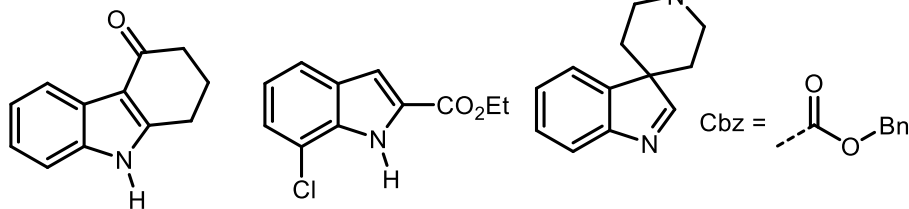
Indol-Synthese (1883/84)



Emil Fischer

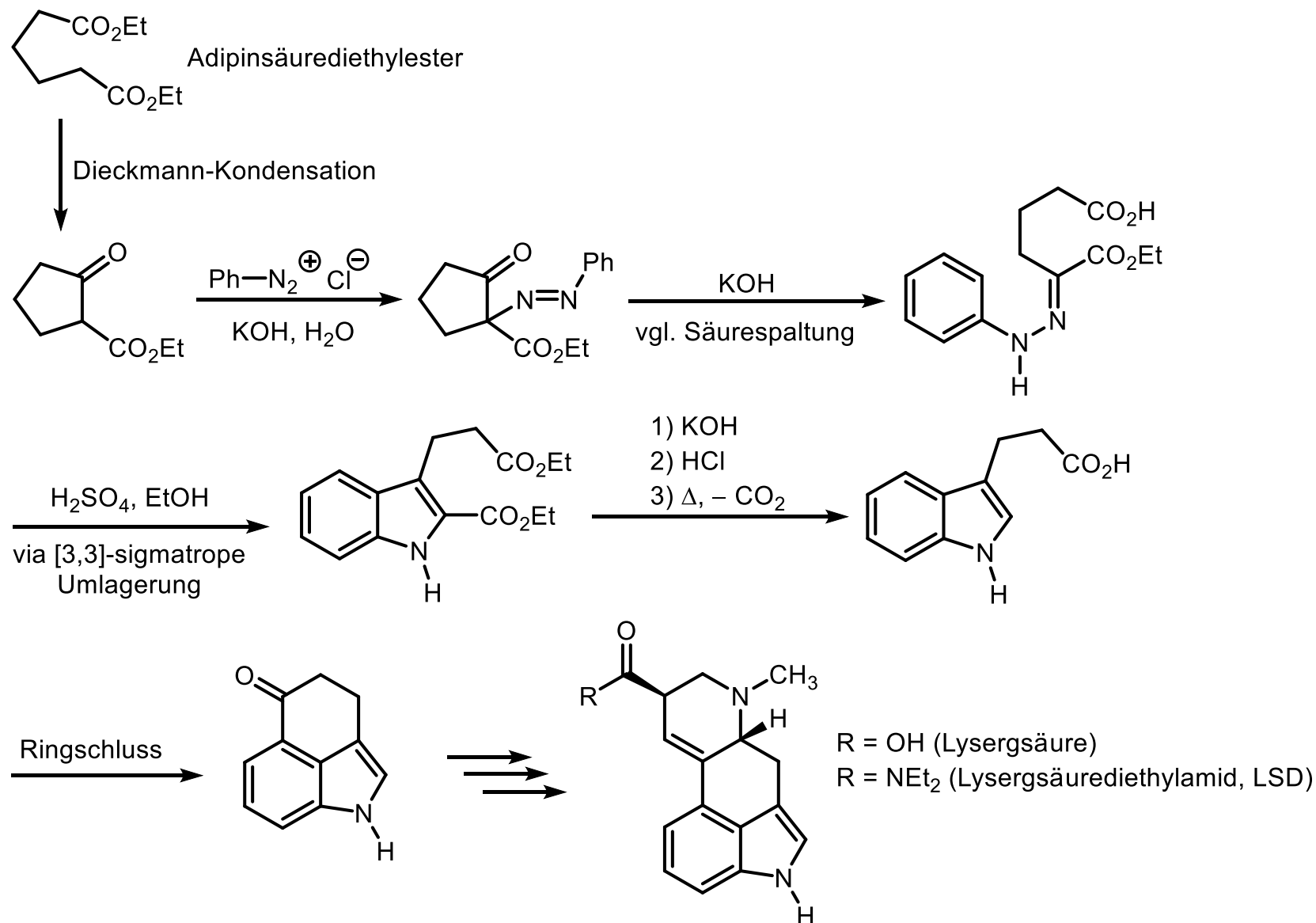
Emil Fischer
(1852-1919)

Übungsaufgaben



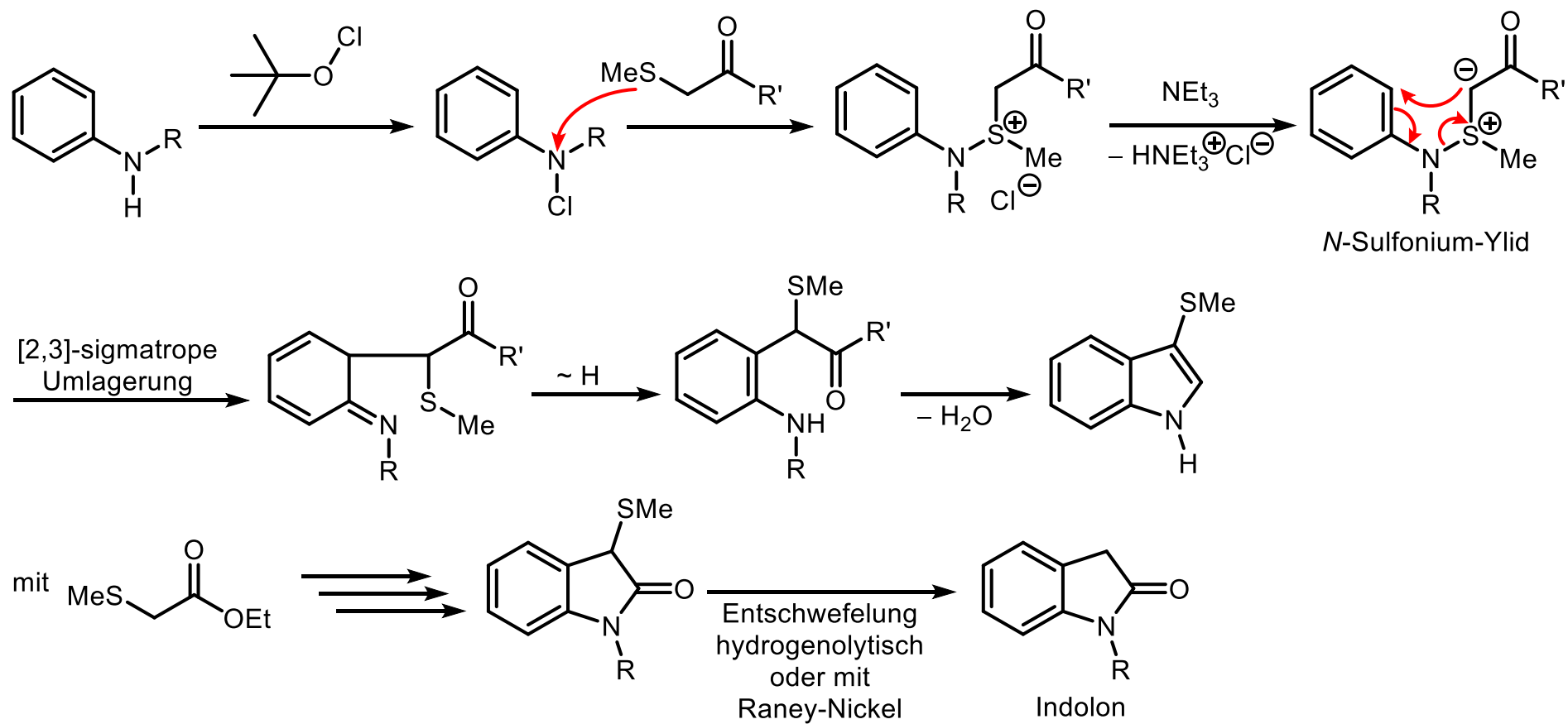
4.5.4. Ringaufbauende Synthesen

Japp-Klingemann-Reaktion zur Synthese von Phenylhydrazonen



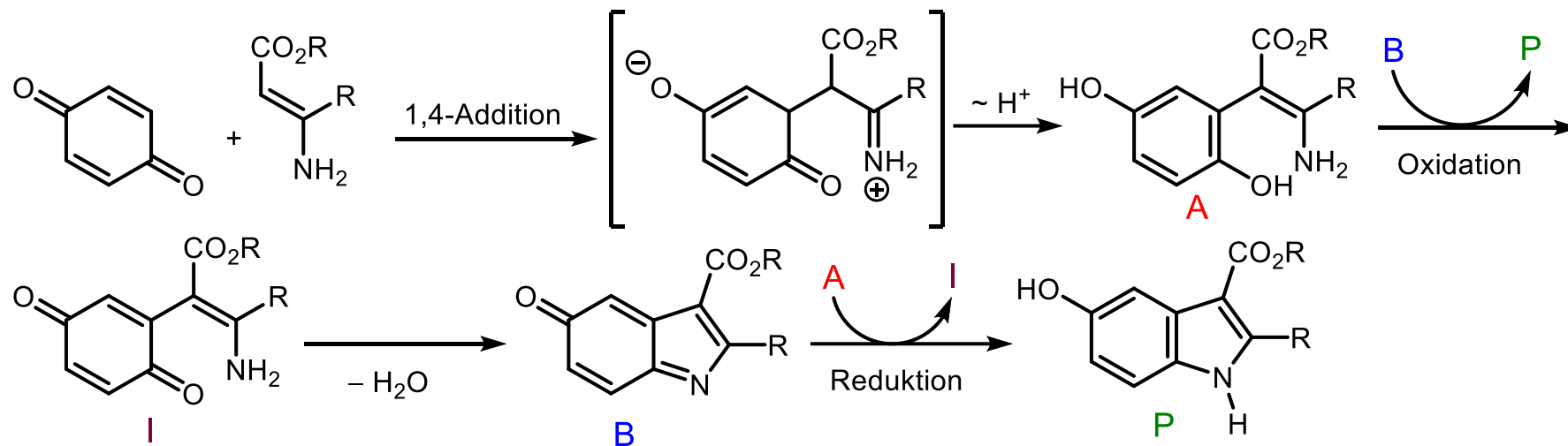
4.5.4. Ringaufbauende Synthesen

Gassman-Synthese



4.5.4. Ringaufbauende Synthesen

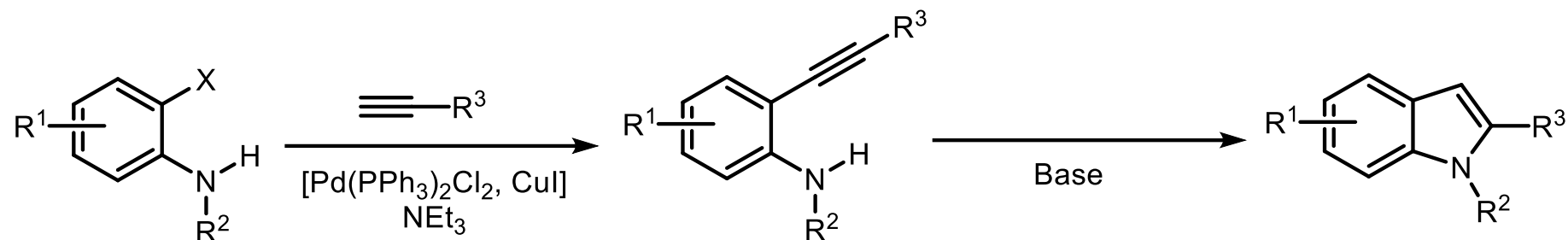
Nenitzescu-Synthese



Wechselseitige Oxidation/Reduktion der Intermediate **A** und **B** zu **I** und **P**

4.5.4. Ringaufbauende Synthesen

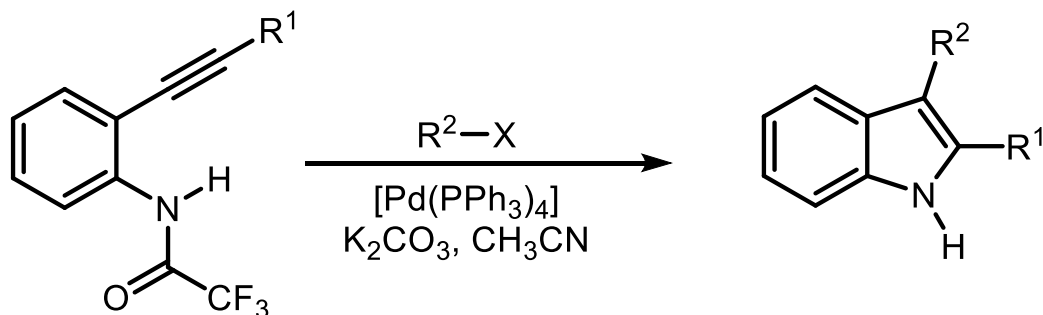
Arcadi-Cacchi-Synthese (Sonogashira-Kreuzkupplung und Castro-Indolsynthese)



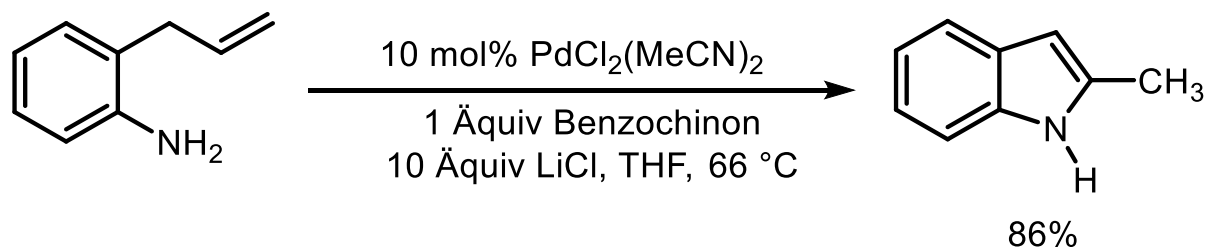
$R^2 = \text{H, COR}', \text{SO}_2\text{R}'$

$X = \text{I, Br}$

- Substituenten-abhängiges Gemisch aus o-Alkinylanilin und Indol
- Cyclisierung zum Indol auch in alleiniger Anwesenheit von Pd oder Cu



Hegedus-Synthese (ausgehend von o-Allylanilinen)



Arcadi-Cacchi-Synthese (Festphasensynthese)

Reaction scheme showing the conversion of a diol to a chloromethyl ether:

Starting material: 4-(4-chloromethylphenoxy)-4'-methylbiphenyl-2-ol

Reagents: MsCl, LiCl, Collidin, DMF

Product: 4-(4-chloromethylphenoxy)-4'-methylbiphenyl-2-yl methanesulfonylate (represented as a cationic species with a positive charge on the phosphorus atom of the mesylate group).

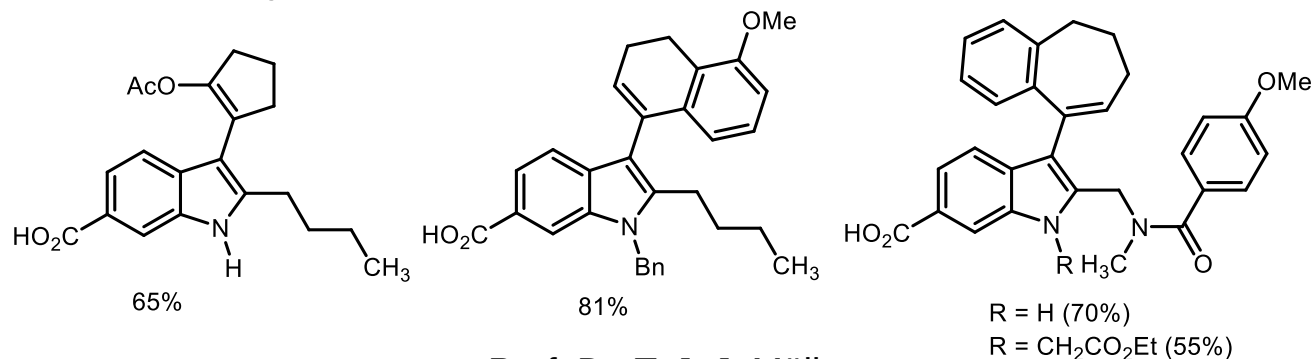
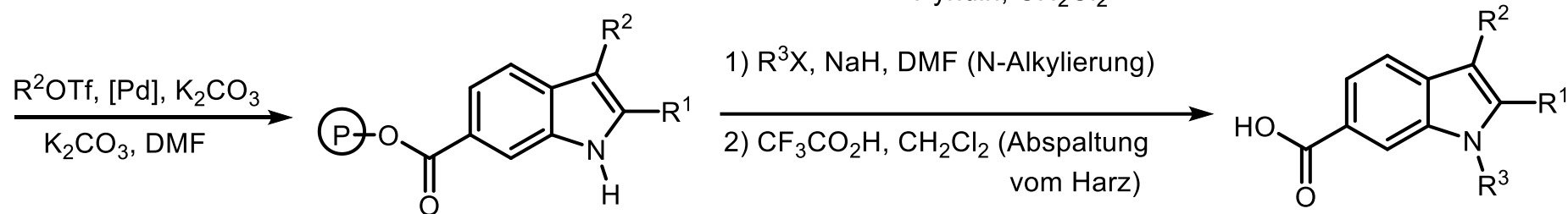
Wang-Harz (Wang resin)

Reaction scheme showing the synthesis of a resin-bound alkyne derivative from a Wang-Harz resin starting material.

Starting material: 2-iodo-4-(carbamoyloxy)benzoic acid (HO₂C-C₆H₃(I)-NH₂).

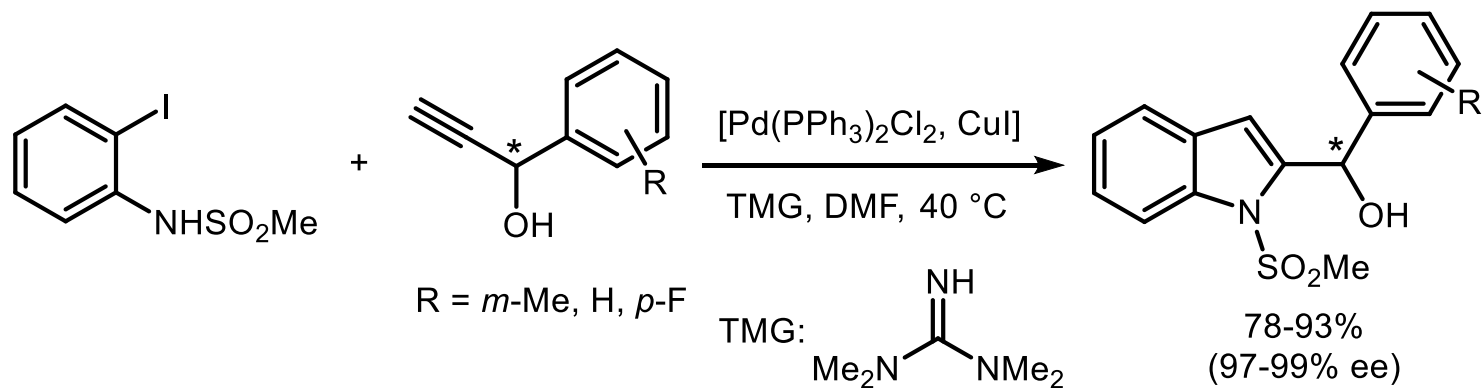
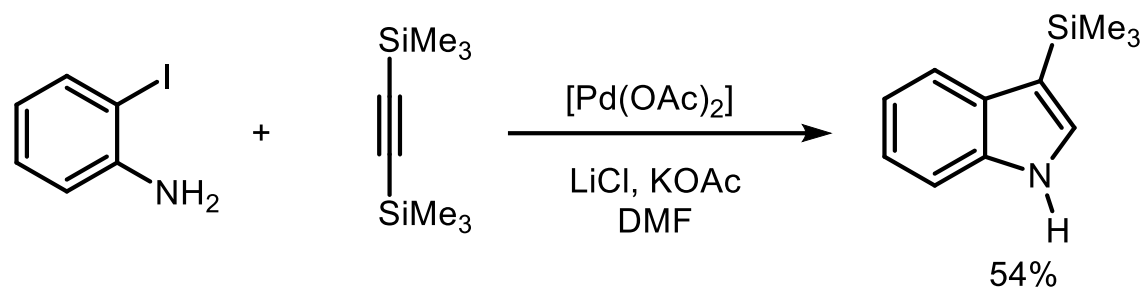
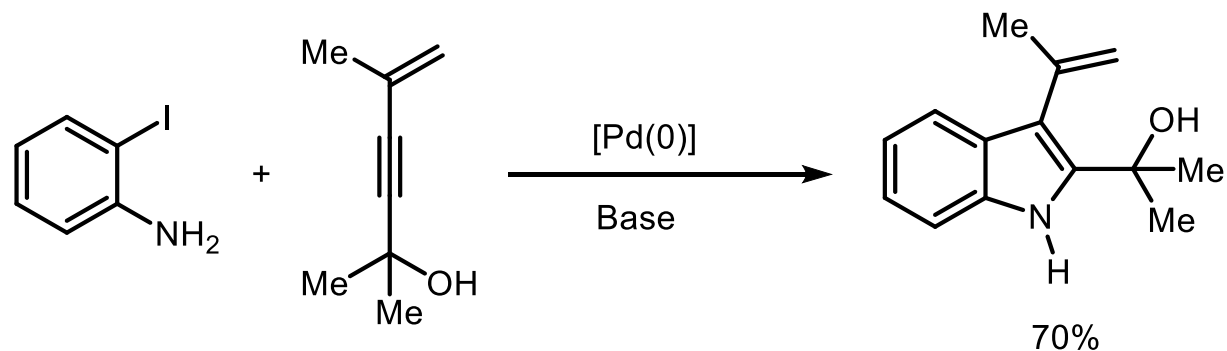
Step 1: Reaction with a resin-bound phosphonate (P-Cl) in the presence of Cs₂CO₃ and DMF yields a resin-bound intermediate (P-O-CO-C₆H₃(I)-NH₂).

Step 2: Reaction with an alkyne (R¹-C≡CH) and [Pd/Cu], Et₂NH, DMF, followed by (CF₃CO)₂O, Pyridin, CH₂Cl₂, yields the final resin-bound alkyne derivative (P-O-CO-C₆H₃(C≡C-R¹)-NHCOCF₃).



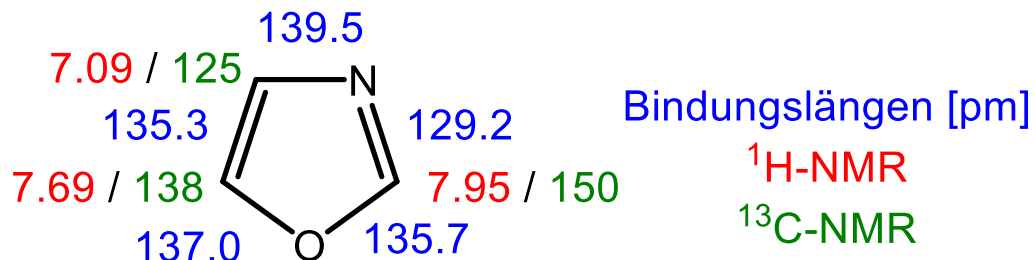
4.5.4. Ringaufbauende Synthesen

Larock-Synthese



4.6. Oxazol

4.6.1. Struktur, Eigenschaften

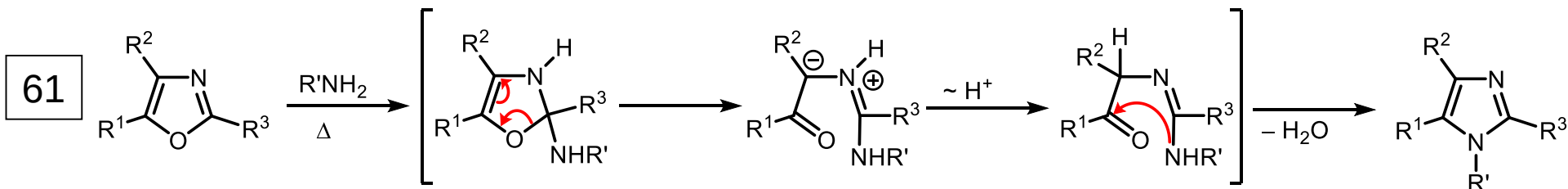


Sdp. 70 °C

4.6.2. Reaktivität

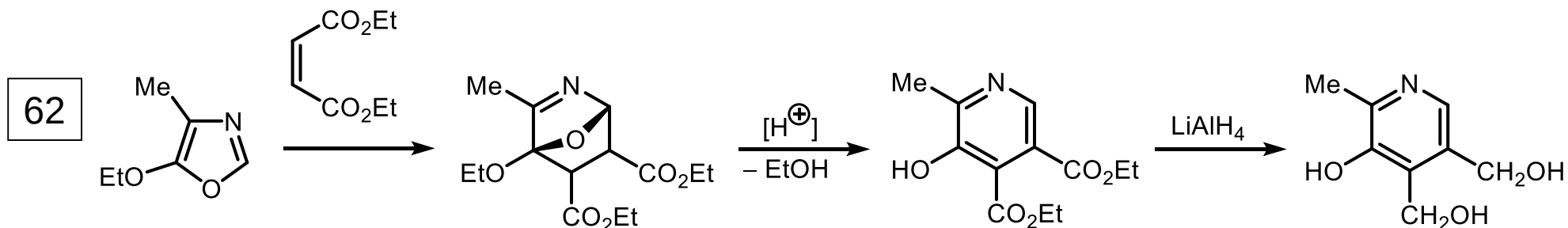
- vergleichbar mit Furan; elektronenärmste Position C₂.
- Lithiierung an C₂ (+ DMF -> 2-Formylierung).
- Quaternierung mit Haloalkanen am N.
- Synthese-Intermediat auf dem Weg zu anderen Heterocyclen

a) nucleophiler Angriff an C₂: Umwandlung in Imidazol

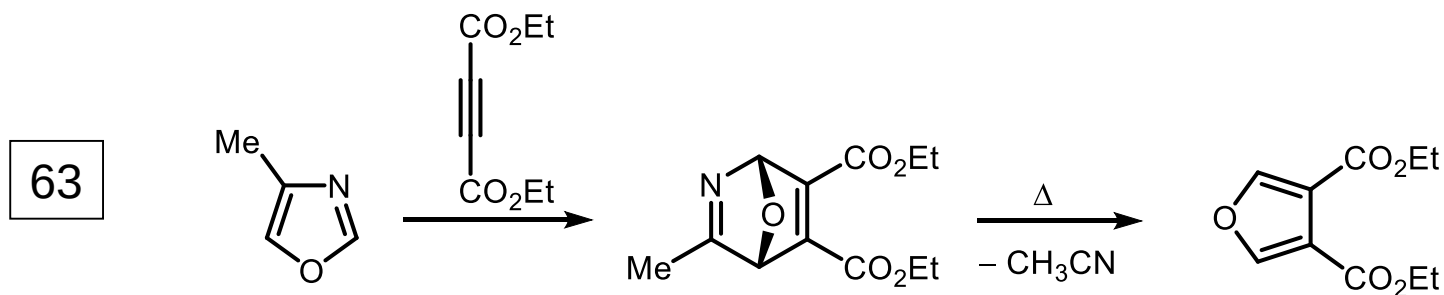


b) Diels-Alder-Reaktionen

Transformation zu Pyridin:



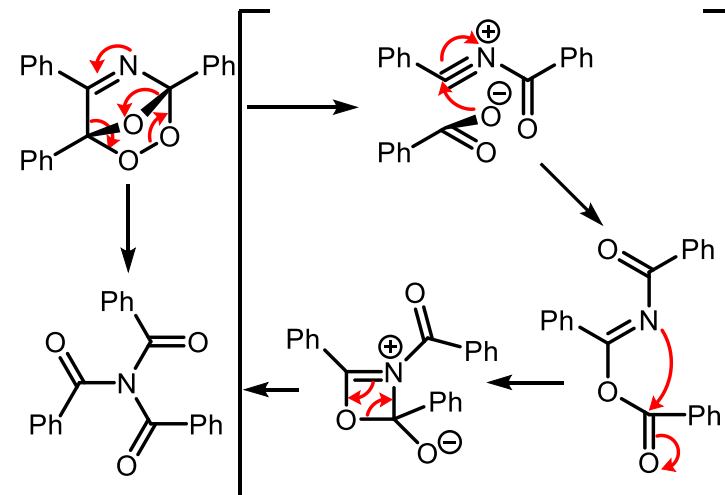
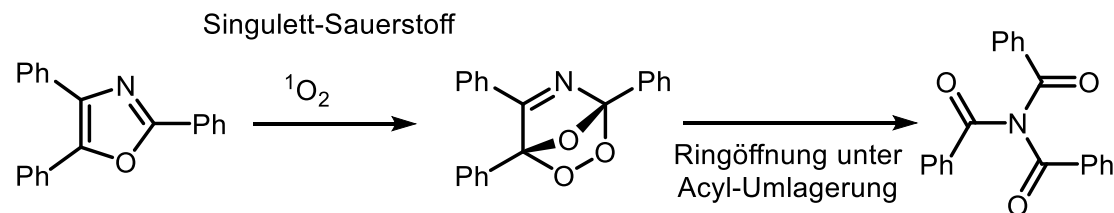
Transformation zu Furan:



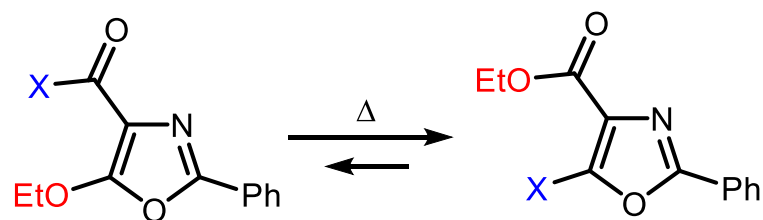
4.6.2. Reaktivität

Fortsetzung

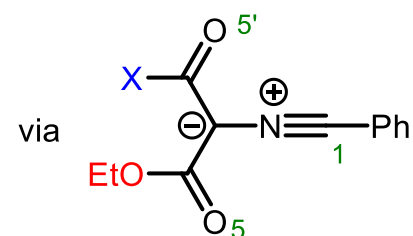
Reaktion mit Singulett-Sauerstoff:



c) Cornforth-Umlagerung



X = $-\text{NR}_2$, Pyrazolyl-, Imidazolyl



Nitrilylid-Zwischenstufe
(Bildung/Reaktion über 1,5-dipolare Cyclisierung
6 π -elektrocyclische Reaktion)

4.6.3. Ringaufbauende Synthesen

a) Robinson-Gabriel-Synthese

